Highlights

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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- Transfer-function contrast, not a single scenario ratio, explains vibration–sound asymmetry
- Oblate spheroidal shell model predicts abdominal resonance at 4–10 Hz
- Breathing mode at ∼2500 Hz; flexural modes at infrasonic frequencies
- Airborne coupling penalised by $(ka)^n$; negligible at 120 dB SPL
- Monte Carlo UQ shows the transfer-function contrast remains decisive; Sobol $S_T = 0.86$ for modulus

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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Abstract

Whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal disturbances, yet airborne sound at the same frequencies does not—even at extreme intensities. This asymmetry underlies the persistent "brown note" hypothesis: the conjecture that infrasonic sound can induce involuntary defecation. We resolve the asymmetry by modelling the human abdomen as a fluid-filled viscoelastic oblate spheroidal shell and quantifying both excitation pathways within a common energy-budget framework.

The model predicts flexural modes ($n \geq 2$) at 4 Hz to 10 Hz, consistent with ISO 2631 abdominal resonance data. At these frequencies the abdomen lies deep in the Rayleigh regime ($ka \approx 0.01$); the resulting $(ka)^n$ spatial-filtering penalty suppresses modal-pressure content by four orders of magnitude, while the air–tissue impedance mismatch limits energy transfer to the shell. The corresponding airborne and mechanical transfer functions at the $n = 2$ resonance are $H_{\mathrm{air}} \approx 1.4 \times 10^{-9}\,\mathrm{m/Pa}$ and $H_{\mathrm{mech}} \approx 9.17 \times 10^{-3}\,\mathrm{s}^2$ (SDOF upper bound, before modal-participation correction), giving a canonical transfer-function ratio $H_{\mathrm{mech}}/H_{\mathrm{air}} \approx 6.5 \times 10^6\,\mathrm{Pa\,s^2/m}$. Energy-consistent analysis yields wall displacements of 0.028 µm at 120 dB SPL—far below cellular mechanotransduction benchmarks. For the reference exposure pair 120 dB SPL and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$, this transfer-function contrast implies a displacement ratio of approximately 3.3×10^4. Monte Carlo analysis spanning representative anatomical variability

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confirms that the transfer-function contrast remains decisive throughout the parameter space, with the fundamental flexural frequency varying between 3 and 16 Hz (90% credible interval, median 6.4 Hz; $S_T = 0.86$ for elastic modulus). The global abdominal-cavity resonance is physically real, but airborne sound at any plausible intensity cannot excite it to physiologically relevant amplitudes.
Keywords: fluid-filled shell vibration, viscoelastic oblate spheroid, infrasound, whole-body vibration, acoustic–structural coupling, ISO 2631

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1. Introduction

Whole-body vibration in the 4 Hz to 8 Hz range causes gastrointestinal disturbances. This is an established occupational-health effect, codified in ISO 2631 [1], reviewed by Griffin [2] and Mansfield [3], and documented among truck drivers, helicopter pilots, and heavy-equipment operators worldwide. Airborne sound at the same frequencies does not produce these effects, even at extreme intensities. NASA-funded exposures at 140–154 dB SPL caused chest-wall vibration and respiratory disruption, but *no gastrointestinal response* [4]. Two excitation pathways, the same target organ, the same frequency band—yet outcomes that differ by orders of magnitude. This contrast motivates the present study.

The "brown note" hypothesis—a hypothetical infrasonic tone alleged to induce involuntary bowel evacuation—has also circulated in military folklore, audio engineering, and popular media. Scientifically, the claim reflects a persistent confusion between what mechanical vibration demonstrably does and what airborne sound is assumed to do. A more plausible airborne route, insofar as one exists, is vestibular rather than visceral: recent measurements show that infrasound can be transmitted through the ear strongly enough to engage acoustic and vestibular pathways long before any whole-cavity abdominal resonance is appreciably driven [5].

The empirical asymmetry between mechanical and airborne excitation is, in fact, a straightforward consequence of long-wavelength acoustics. At the n=2 resonance ($f_2 \approx 4$ Hz) the acoustic wavelength exceeds 85 m—more than 250

times the abdominal diameter. In this deep Rayleigh regime, the n-th spatial harmonic of the incident field scales as $(ka)^n$, where $ka \approx 0.01$. Flexural deformation of the abdominal wall ($n \geq 2$) therefore receives a driving pressure four orders of magnitude below the free-field value. The air–tissue impedance mismatch compounds the deficit from the energy side: at the interface, approximately 99.9 % of incident acoustic power is reflected even before the much smaller mode-specific resonant absorption is considered, so the energy available to drive flexural modes is negligible. Mechanical vibration, transmitted through the skeleton and abdominal wall, suffers neither penalty. The resulting separation between the airborne and mechanical transfer functions is the central system property we derive and the principal contribution of this paper.

Scientific discussion of infrasound predates later popular-media portrayals, including the *South Park* episode "World Wide Recorder Concert" (also known as "The Brown Noise") [6]. Gavreau's experiments at CNRS Levallois-Perret in the late 1950s, where a faulty ventilator at approximately 7 Hz caused nausea among laboratory staff [7, 8], drew early attention to infrasound. Von Gierke's systematic review for NASA [9, 10] established exposure criteria that informed later ISO standards. Tandy and Lawrence [11] demonstrated that a 19 Hz standing wave could induce visual disturbances and anxiety—real perceptual effects, but far from the claimed bowel response. Critical reviews by Leventhall [12, 8] and assessments of acoustic weapons by Altmann [13] and Jauchem and Cook [14] concluded that infrasound is impractical as a weapon and that claims of extreme physiological effects are exaggerated or fabricated. Recent wind-turbine reviews likewise find that the dominant health concerns are annoyance, sleep disturbance, and other indirect effects, not visceral resonance [15]. Randomised exposure studies with neuroimaging have also reported subtle but measurable neural correlates of airborne infrasound exposure, but again no evidence of gastrointestinal or abdominal-resonance effects [16]. Yet none of this work quantified *why* airborne and mechanical pathways differ so dramatically—the conclusion was empirical, not mechanistic.

We provide the mechanistic account. To our knowledge, no prior analytical

study has combined first-principles cavity–shell modal analysis with an energy-consistent comparison of airborne and mechanical coupling pathways:

1. derived abdominal cavity eigenfrequencies from first-principles shell theory with proper fluid–structure coupling;
2. quantified the airborne acoustic coupling coefficient for flexural modes in the long-wavelength limit; or
3. compared airborne and mechanical excitation pathways within a common energy-budget framework to obtain the governing transfer functions and the exposure-pair displacement ratios derived from them.

We address all three gaps. We model the abdomen as a fluid-filled viscoelastic oblate spheroidal shell, derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families, and compare airborne acoustic coupling—subject to a severe $(ka)^n$ penalty at infrasonic wavelengths—against direct mechanical (whole-body) vibration coupling, which suffers no such penalty. More broadly, this provides a general template for comparing excitation pathways in acoustically compact, fluid-loaded compliant structures. The same framework identifies when geometry, impedance mismatch, and modal participation make one forcing route dominant over another even when both act on the same target mode.

The headline result is therefore not a single scenario-specific number but a large separation between transfer functions. At the canonical $n = 2$ resonance, $H_{\mathrm{mech}}/H_{\mathrm{air}} \approx 6.5 \times 10^6\,\mathrm{Pa\,s^2/m}$ before the modal-participation correction, and the corresponding reference-pair displacement ratio for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration is approximately 3.3×10^4 under the SDOF idealisation ($\Gamma_2 = 1$). Including the canonical participation factor $\Gamma_2 \approx 0.48$ reduces this to $\sim 1.6 \times 10^4$. Monte Carlo uncertainty quantification confirms that the transfer-function contrast persists across physiologically realistic parameter ranges and resolves the empirical asymmetry cleanly. It explains why whole-body vibration at 4 Hz to 8 Hz produces well-documented gastrointestinal effects, why airborne infrasound at comparable frequencies does

not, and why claims of a brown note are not supported by the physics of airborne acoustic coupling.

The remainder of this paper is organised as follows. Section 2 develops the mathematical formulation. Section 3 presents a parametric study and compares the model against ISO 2631 data. Section 4 quantifies the coupling disparity. Section 5 discusses implications, alternative pathways, and limitations, and Section 6 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human abdominal cavity is idealised as an oblate spheroidal shell of semi-major axes a (equatorial, in the lateral and cranio-caudal directions) and semi-minor axis c (antero-posterior), with $c < a$. The shell has uniform wall thickness h and is completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Following standard practice in fluid-filled shell analysis [17, 18] and recent experimental and analytical studies of pressurised elastic shells [19, 20], an equivalent-sphere radius is defined via

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which preserves the cavity volume, $V = \frac{4}{3}\pi a^2 c$. This approximation is exact for a sphere ($c = a$) and introduces errors of order $(1-c/a)^2$ for the eigenfrequencies at moderate aspect ratios ($c/a \gtrsim 0.5$). Table 1 lists the baseline geometric and material parameters with physiological justification.

The thin-shell assumption ($h/R_{\mathrm{eq}} \ll 1$) is satisfied: for $h = 10\,\mathrm{mm}$ and $R_{\mathrm{eq}} \approx 157\,\mathrm{mm}$, $h/R \approx 0.06$. The Love–Kirchhoff hypothesis [21] (normals remain straight and normal to the deformed mid-surface) is therefore applicable. Classical thin-shell theory is used for analytical tractability; higher-order shear effects are expected to perturb frequencies modestly rather than overturn the coupling argument. This linear framework remains the standard starting point even in modern reviews of nonlinear shell vibration, which treat geometric nonlinearities as amplitude corrections to the same modal backbone [22].

Table 1: Baseline model parameters with physiological justification.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Elastic modulus	E	0.1	MPa	Representative of reported measurements [23]
Poisson's ratio	ν	0.45	—	Near-incompressible modelling convention [24]
Wall density	ρ_w	1100	kg/m^3	Soft tissue
Fluid density	ρ_f	1020	kg/m^3	Abdominal fluid
Fluid bulk modulus	K_f	2.2	GPa	Water at 37°C
Intra-abdominal pressure	P_{iap}	1000	Pa	Representative resting clinical measurement [25]
Loss tangent	η	0.25	—	Representative review value [26]

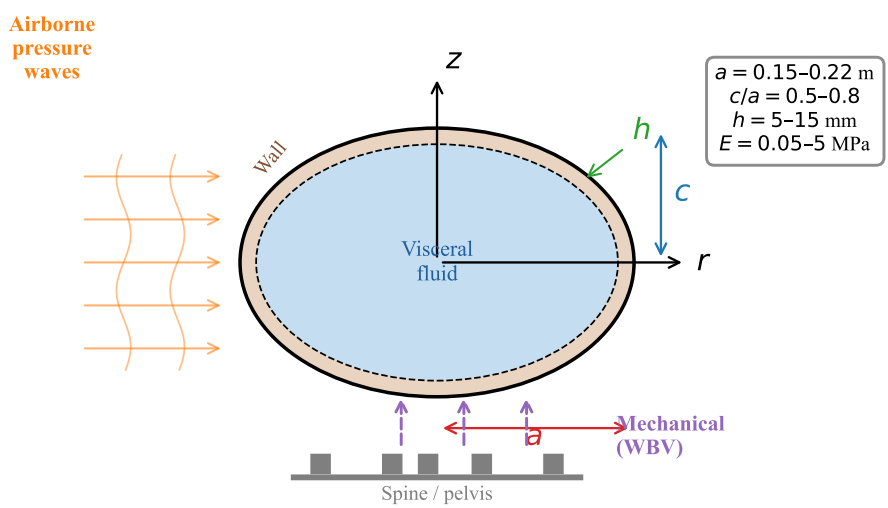

Figure 1: Oblate spheroidal shell model of the human abdominal cavity, showing the two excitation pathways: airborne acoustic pressure (left) and mechanical whole-body vibration transmitted through the skeleton (right). Dimensions correspond to the baseline parameters in Table 1.

Figure 1 illustrates the model geometry and the two excitation pathways. The abdominal wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{2}$$

where η is the loss tangent. For small damping, the quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2 = 1/(2Q)$. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1-\nu^2)}. \tag{3}$$

These baseline values are not arbitrary. The canonical modulus $E = 0.1\,\mathrm{MPa}$ is representative of direct abdominal-wall measurements compiled by Deeken and Lake [23]; $\nu = 0.45$ follows the nearly incompressible modelling convention adopted in recent abdominal-wall finite-element studies [24]; resting intra-abdominal pressures of 5–12 mmHg correspond to approximately 0.67 kPa to 1.60 kPa, so $P_{\mathrm{iap}} = 1.0\,\mathrm{kPa}$ is representative of direct clinical measurements [25]; and modern elastography reviews place the soft-tissue loss tangent broadly in the 0.1–0.4 range, making $\eta = 0.25$ consistent with the range reported for soft tissue [26]. Published values of η for abdominal soft tissue range from 0.15 to 0.40 at low frequencies [27], corresponding to $Q = 2.5$–6.7. The baseline value $\eta = 0.25$ ($Q = 4.0$, $\zeta = 0.125$) represents a central estimate.

2.2. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation in which the shell displaces by ξ at every point. The associated volumetric strain is $\Delta V/V = 3\xi/R$, which compresses the contained fluid. Two restoring forces act on the shell:

Membrane stiffness.. For the uniform expansion of a thin spherical shell,

$$k_{\mathrm{shell}} = \frac{2Eh}{R^2(1-\nu)}. \tag{4}$$

Fluid volumetric stiffness.. The pressure increment due to fluid compression is $\Delta p = K_f \Delta V/V = 3K_f\xi/R$. The restoring force per unit displacement per unit

area is

$$k_{\text{fluid}} = \frac{3K_f}{R}. \tag{5}$$

With $K_f = 2.2\,\text{GPa}$ and $R = 0.157\,\text{m}$, $k_{\text{fluid}} = 4.2 \times 10^{10}\,\text{Pa/m}$, which exceeds the membrane stiffness ($k_{\text{shell}} \approx 1.5 \times 10^5\,\text{Pa/m}$) by five orders of magnitude. *The breathing mode is entirely dominated by the fluid bulk modulus.*

The effective mass per unit area includes the shell wall and the fluid added mass for the $n = 0$ mode:

$$m_{\text{eff}}^{(0)} = \rho_w h + \rho_f R. \tag{6}$$

The breathing mode frequency is therefore

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}} \approx 2500\ \text{Hz}, \tag{7}$$

which lies in the kilohertz range and is irrelevant to infrasonic exposure. This result is robust to material property variations because $k_{\text{fluid}} \gg k_{\text{shell}}$ for any realistic shell stiffness.

2.3. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes (oblate–prolate oscillation for $n = 2$, trilobed for $n = 3$, etc.) that preserve the enclosed volume to leading order. The $n = 1$ mode corresponds to rigid-body translation and therefore has zero natural frequency for a free shell, so it is excluded from the resonance discussion. The $n = 2$ mode is consequently the lowest physical flexural mode and the primary candidate for abdominal resonance; higher modes ($n = 3, 4, \ldots$) lie at progressively higher frequencies and are retained for sensitivity studies, but the $n = 2$ family dominates the physiologically relevant 4 Hz to 8 Hz band. The fluid therefore does *not* act as a volumetric spring; it contributes only added mass.

For a partially constrained shell—as when the pelvis and spine anchor the inferior portion—the $n = 1$ mode ceases to be a pure rigid-body translation and becomes a support-dominated "belly bounce" with a frequency determined

primarily by the attachment stiffness rather than by shell elasticity. Because this frequency depends on the mechanical properties of the pelvic support (which are not modelled here), the present analysis excludes it and focuses on $n \geq 2$ deformational modes whose frequencies are governed by the shell–fluid system itself. We note, however, that the existence of a constrained $n = 1$ mode would strengthen, not weaken, the case for mechanical coupling: it would provide an additional low-frequency pathway for whole-body vibration, while the airborne suppression at $ka \ll 1$ would remain equally severe for $n = 1$ as for $n = 2$.

Following Lamb [18] and Junger & Feit [17], with subsequent verification in modern pressurised-shell studies [19, 20], the flexural mode of order n has three contributions to the restoring force per unit area. (An analogous modal decomposition for fluid-filled elastic tubes in biological contexts was developed by Heil [28].)

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{8}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}. \tag{9}$$

Pre-stress stiffness.. The static intra-abdominal pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{10}$$

The fluid added mass per unit area for mode n is

$$m_{\text{add}}^{(n)} = \frac{\rho_f R}{n}, \tag{11}$$

giving a total effective mass of

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{12}$$

The natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}. \tag{13}$$

For the baseline parameters (Table 1), $f_2 = 4.0\,\mathrm{Hz}$, $f_3 = 6.3\,\mathrm{Hz}$, and $f_4 = 8.9\,\mathrm{Hz}$ (Figure 2). These lie squarely in the 4–10 Hz range reported for abdominal resonance in ISO 2631 [1] and in the experimental studies of Kitazaki & Griffin [29] and Mansfield [3].

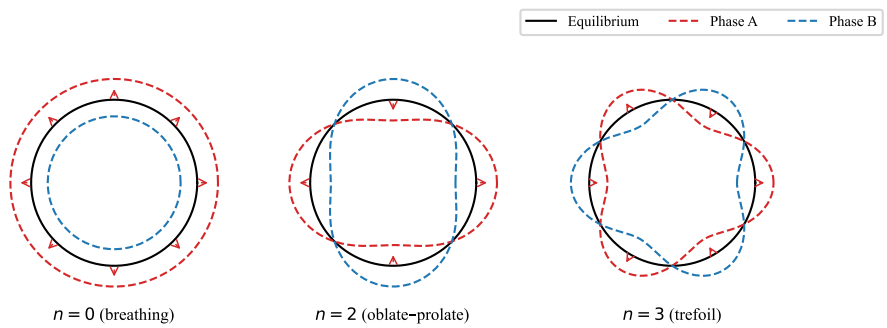

Figure 2: Schematic cross-sections of the first three mode families. The breathing mode ($n = 0$) involves uniform radial expansion against the fluid bulk modulus. Flexural modes ($n = 2, 3$) deform the shell shape without significant volume change; the enclosed fluid acts as added mass only.

2.4. Airborne Acoustic Coupling

We now quantify the excitation of flexural modes by an incident airborne acoustic field. At $f_2 \approx 4\,\mathrm{Hz}$, the acoustic wavelength in air is $\lambda \approx 87\,\mathrm{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R_{\mathrm{eq}}}{c_{\mathrm{air}}} \approx 0.0114 \ll 1. \tag{14}$$

The body is deep in the sub-wavelength (Rayleigh scattering) regime.

In a multipole expansion of the incident plane wave scattered by a sphere, the n-th order component of the pressure field scales, to leading order, as [17]:

$$p_{\mathrm{eff}}^{(n)} \propto p_{\mathrm{inc}}(ka)^n. \tag{15}$$

The proportionality constant from the multipole expansion is $(2n+1)/(2n+1)!!$ [17], which evaluates to $1/3$ for $n = 2$. In what follows, we set this prefactor to unity as a deliberate upper bound on the modal pressure; the exact coefficient reduces the pressure-based displacement estimate by a factor of three.

The breathing mode ($n = 0$) sees the full incident pressure, but flexural modes ($n \geq 2$) see only the higher-order spatial gradients, which are vanishingly small when $ka \ll 1$. For $n = 2$:

$$p_{\text{eff}}^{(2)} \propto p_{\text{inc}} \, (ka)^2 \sim p_{\text{inc}} \times 1.3 \times 10^{-4}, \tag{16}$$

where the upper-bound convention (unity prefactor) has been adopted; the exact multipole coefficient of 1/3 would reduce this estimate by a factor of three.

The displacement at resonance ($\omega = \omega_n$) is

$$\xi_{\text{air}} = \frac{p_{\text{eff}}^{(n)}}{K_{\text{total}}^{(n)}} Q, \tag{17}$$

where $K_{\text{total}}^{(n)} = K_{\text{bend}} + K_{\text{memb}} + K_P$. To place this displacement in biological context, we compare it with cellular mechanotransduction benchmarks. PIEZO1 and PIEZO2 are mechanically activated ion channels that convert membrane deformation into cellular signals; reported activation in cell-culture experiments begins for deformations of order 0.5 µm to 2.0 µm [30]. While these cell-scale values cannot be directly translated to tissue-scale abdominal wall displacements, they provide a useful order-of-magnitude reference. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$), the energy-consistent reciprocity analysis (derived in §4.4) gives an energy-consistent wall displacement of approximately $0.028\,\mu$m for $n = 2$ — well below the 0.5 µm to 2.0 µm cellular activation benchmark. (The pressure-based upper-bound modal-displacement estimate from equation (17) yields approximately $0.18\,\mu$m, an overestimate that violates the energy budget; see §4.4.)

2.5. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration (WBV) transmits through the skeleton and acts as *base excitation* of the abdominal shell. Unlike airborne acoustic coupling, there is no $(ka)^n$ penalty: the impedance mismatch between bone and peritoneal wall is orders of magnitude smaller than the air–tissue contrast, and the vibration bypasses the geometric filtering entirely. The following analysis therefore represents an *upper-bound* base-excitation model in which the skeletal frame transmits vibration rigidly to the shell boundary.

For harmonic base acceleration $a_{\text{base}}(t) = a_0 \cos(\omega t)$, the base displacement amplitude is

$$x_{\text{base}} = \frac{a_{\text{rms}}\sqrt{2}}{\omega^2}. \tag{18}$$

The relative displacement between the shell and the skeleton is given by the base-excitation frequency response function:

$$H_{\text{rel}}(\omega) = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_n}, \tag{19}$$

so that

$$\xi_{\text{mech}} = x_{\text{base}} \times H_{\text{rel}}. \tag{20}$$

At the EU Whole-Body Vibration Directive exposure limit of $a_{\text{rms}} = 1.15\,\text{m/s}^2$ and at the $n = 2$ resonance frequency:

$$\xi_{\text{mech}} \approx 10{,}500\ \mu\text{m} \approx 10\ \text{mm}, \tag{21}$$

which exceeds the cellular mechanotransduction benchmark by more than three orders of magnitude.

2.6. Coupling Ratio

The ratio of mechanical to airborne displacement at the same excitation conditions provides a dimensionless measure for any chosen exposure pair:

$$\mathcal{R}_{\text{full}} = \frac{\xi_{\text{mech}}}{\xi_{\text{air}}}. \tag{22}$$

This ratio depends primarily on ka and the excitation level, and the qualitative ordering persists across plausible parameter ranges because both pathways excite the same modal structure. Section 4 shows that the underlying system property is the large separation between the transfer functions H_{mech} and H_{air}; the corresponding dimensionless ratio then follows once a reference pressure and acceleration are chosen. For 120 dB SPL airborne excitation and $a_{\text{rms}} = 0.1\,\text{m/s}^2$ RMS whole-body vibration, the resulting SDOF upper bound is $\mathcal{R}_{\text{full}} \approx 3.3\times10^4$ ($\Gamma_2 = 1$). However, vertical base excitation does not couple perfectly into the $n = 2$ mode: the modal participation factor $\Gamma_2 \approx 0.48$ (§4.3) reduces the effective mechanical displacement to about half the SDOF value, giving a corrected

ratio of $\sim 1.6 \times 10^4$. Even with this correction, the four-order-of-magnitude gap between pathways is unaffected.

3. Parametric Study

3.1. Elastic Modulus Sensitivity

The $n = 2$ flexural mode frequency depends strongly on the wall elastic modulus E. Table 2 shows the predicted frequency for a free shell across the physiologically relevant range. The sweep 0.05 MPa to 2.0 MPa spans reported values from passive peritoneum and relaxed abdominal wall tissue to tensed abdominal musculature [23, 24, 26, 27].

Table 2: Flexural mode $n = 2$ frequency vs. elastic modulus ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$). The range 0.05 MPa to 2.0 MPa is intended to bracket passive to active abdominal-wall stiffness reported in the abdominal mechanics and elastography literature [23, 24, 26, 27].

E (MPa)	f_2 (Hz)	In ISO 2631 range?
0.05	3.4	Marginal
0.10	4.0	Yes
0.20	4.9	Yes
0.50	7.1	Yes
1.00	9.6	Marginal
2.00	13.3	No

The 4 Hz to 8 Hz ISO 2631 abdominal resonance range requires $E < 0.5$ MPa, corresponding to relaxed, passive musculature. Active muscle bracing increases E by a factor of 10–100, moving the resonance well above the infrasound range. Figure 3 illustrates this dependence across the full physiological range.

3.2. Multi-Parameter Sensitivity

A 486-point full factorial analysis varying E, a, c/a, h, and η reveals that 37% of configurations place the $n = 2$ mode in the 5 Hz to 10 Hz range and 41% in

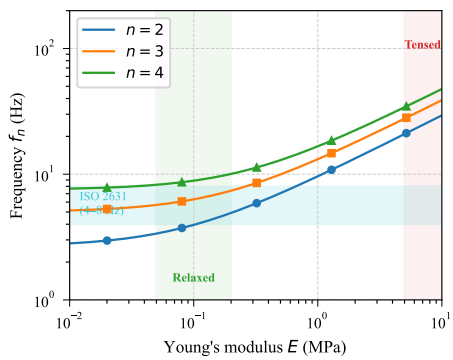

Figure 3: Flexural mode frequencies vs. elastic modulus for modes $n = 2, 3, 4$. The shaded band marks the ISO 2631 abdominal resonance range (4 Hz to 8 Hz). Relaxed musculature places the $n = 2$ mode within this band; tensed muscle shifts it well above.

the broader 4 Hz to 8 Hz ISO 2631 range. The separate Sobol sensitivity analysis below confirms that E dominates, followed by geometry and intra-abdominal pressure.

3.3. Uncertainty Quantification

A Monte Carlo simulation with $N = 10{,}000$ samples propagates parameter uncertainties through the model. Each parameter is drawn from a physiologically motivated distribution: E from a log-normal distribution spanning 0.05 MPa to 2.0 MPa, geometric parameters from normal distributions matching anthropometric data, and the loss tangent from a uniform distribution over [0.10, 0.40]. A small number of Monte Carlo samples (25 of 10,000, or 0.25%) drew nonphysical negative wall thickness from the untruncated normal distribution and were excluded; the reported statistics are computed from the remaining 9,975 samples.

The resulting distribution of f_2 has a median of 6.4 Hz with a 90% credible interval of 3.3 Hz to 15.5 Hz (Table 3). This interval is deliberately conservative: it reflects the simultaneous variation of *all* model parameters across their full physiological ranges and is therefore considerably wider than any single-parameter sweep. The canonical prediction $f_2 = 3.95$ Hz lies within the credible interval

but near its lower bound, corresponding to the soft-tissue, relaxed-musculature baseline; the median is higher because the log-normal modulus distribution samples stiffer configurations more frequently. The distribution is positively skewed, reflecting the log-normal distribution of E.

Table 3: Monte Carlo uncertainty quantification results ($N = 10{,}000$). The airborne displacement and reference-pair displacement ratio $\mathcal{R}_{\text{full}}$ use the energy-consistent formulation at 120 dB; the pressure-based value is retained elsewhere only as a labelled upper bound for comparison. The mechanical row uses the same reference loading as Section 4: $a_{\text{rms}} = 0.1\,\text{m/s}^2$.

Quantity	Median	90% CI	Units
f_2	6.4	[3.3, 15.5]	Hz
ξ_{air} (120 dB, energy-consistent)	0.027	[0.016, 0.059]	μm
ξ_{mech} ($0.1\,\text{m/s}^2$)	360	[59, 1640]	μm
$\mathcal{R}_{\text{full}}$ (120 dB vs. $0.1\,\text{m/s}^2$)	1.3×10^4	$[2.4 \times 10^3, 4.6 \times 10^4]$	—
SPL for PIEZO1/2 activation (energy-consistent, 1 μm)	152	[145, 156]	dB

Sobol sensitivity analysis reveals that the elastic modulus accounts for 86% of the total output variance (total-order index $S_T = 0.86$). All other parameters contribute less than 10% individually. This result has important experimental implications: any future verification effort should prioritise *in vivo* measurement of abdominal wall stiffness, as this single parameter dominates the uncertainty budget (Figure 4). Table 3 now uses the same energy-consistent airborne displacement and the same $120\,\text{dB}/0.1\,\text{m/s}^2$ reference pair as Section 4, so its $\mathcal{R}_{\text{full}}$ row is directly comparable with the canonical SDOF upper bound of 3.3×10^4 quoted there.

3.4. Multi-Layer Wall Model

The abdominal wall comprises six distinct tissue layers: peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. Using classical laminate theory [31], with layer thicknesses and relaxed/tensed tissue moduli drawn from abdominal-wall measurements and soft-tissue elastography reviews [23, 24, 26], the effective composite modulus for a

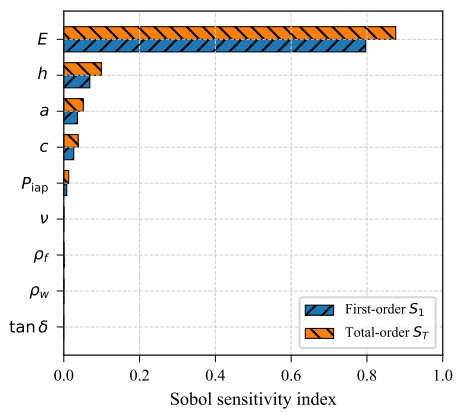

Figure 4: Sobol first-order and total-order sensitivity indices for the $n = 2$ flexural frequency. The elastic modulus E accounts for 86% of the total variance; all other parameters contribute less than 10% individually.

relaxed wall is $E_{\text{eff}} \approx 0.075\,\text{MPa}$ ($h_{\text{total}} = 25.1\,\text{mm}$), giving $f_2 = 4.6\,\text{Hz}$—within approximately 16% of the homogeneous model prediction. Muscle tension is confirmed as the dominant factor: tensed muscle ($E_{\text{eff}} = 2.7\,\text{MPa}$) shifts f_2 to 23.1 Hz.

3.5. Boundary Condition Effects

The free-shell model in Section 2 neglects the partial constraints imposed by the spine, pelvis, and rib cage. We quantify their effect using a Rayleigh–Ritz analysis with Lamb-mode basis functions for four progressively constrained configurations. These configurations represent surrogate constraint cases chosen to bracket the likely effect of anatomical boundary conditions, rather than a strict monotonic continuation of a single geometry.

The Rayleigh–Ritz analysis employs a two-degree-of-freedom trial displacement for each mode order n:

$$w(\eta) = \beta\, P_n(\eta), \qquad u(\theta) = \alpha\, \sin\theta\, P_n'(\cos\theta), \tag{23}$$

where w is the normal (radial) displacement, u is the tangential (meridional) displacement, P_n denotes the Legendre polynomial of degree n, $\eta = \cos\theta$, and α, β

are the Ritz coefficients determined by minimising the Rayleigh quotient. The basis functions are chosen from the Lamb-mode family of free-sphere displacements; for partially constrained configurations, the boundary conditions are enforced weakly through the energy functional rather than strongly at discrete points. In a strict variational setting, such admissible trial functions provide upper bounds on the natural frequencies; in the present reduced two-parameter implementation, the free-shell Ritz frequency remains within 3% of the closed-form Lamb value (Table 5).

Table 5 presents the results. To assess the sensitivity of the modal frequencies to boundary conditions, we performed a comparative analysis using representative alternative parameters ($R = 0.162\,\mathrm{m}$, $\rho_f = 1000\,\mathrm{kg/m^3}$, $P_{\mathrm{iap}} = 1500\,\mathrm{Pa}$) chosen to match a specific mesh geometry; these results are presented for relative comparison rather than absolute prediction. To assess the adequacy of the two-degree-of-freedom basis, we enriched it by adding higher-order Legendre polynomials up to $N = 8$ per displacement component, giving a 16-degree-of-freedom generalised eigenproblem. Table 4 and Figure 5 show that the canonical $n = 2$ flexural frequency converges monotonically from above, consistent with the Ritz upper-bound property. The single-term basis ($N = 1$, 2 DOF) gives $f_2 = 3.7995\,\mathrm{Hz}$, whereas the converged value is $3.6784\,\mathrm{Hz}$; the two-degree-of-freedom model therefore overestimates the converged result by only $3.3\,\%$. Effective convergence is already reached at $N = 3$ (6 DOF), so the reduced basis is adequate for the boundary-condition comparison made here. For the mesh-matched parameter set used in Table 5, the free-sphere Ritz frequency ($f_2 = 4.13\,\mathrm{Hz}$) agrees with the analytical prediction ($4.26\,\mathrm{Hz}$) to within 3%, providing a useful numerical check. The most constrained case—25% of the shell surface clamped—reduces f_2 by only $11.4\,\%$. Even the hemisphere-clamped case shifts the frequency by a mere $5.8\,\%$.

Within this converged Ritz framework, the frequency appears relatively insensitive to boundary conditions. A plausible physical reason is that the fluid added mass for the $n = 2$ mode is $7.4\times$ the shell mass, so the system inertia is dominated by the enclosed fluid rather than the wall. In a dry (empty) shell, the

Table 4: Ritz basis convergence for the $n = 2$ flexural mode.

N	DOFs	f_2 (Hz)
1	2	3.800
2	4	3.682
3	6	3.679
4	8	3.678
8	16	3.678

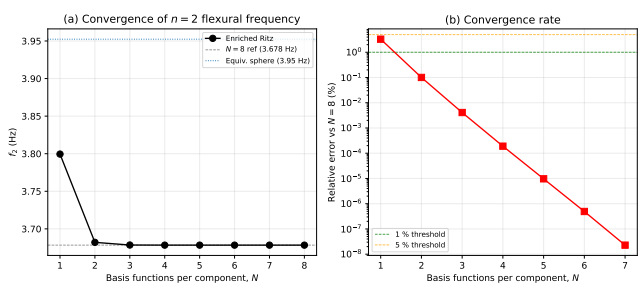

Figure 5: Convergence of f_2 with Ritz basis enrichment. The dashed line marks the converged value; the two-degree-of-freedom result ($N = 1$) lies 3.3 % above it.

Table 5: Rayleigh–Ritz Lamb-mode frequencies under various boundary conditions ($n = 2$ flexural mode).

Configuration	f_2 (Hz)	Shift from free
Free sphere (Ritz)	4.13	0% (verifies analytical)
Hemisphere simply-supported	3.91	−5.3%
Hemisphere clamped	3.89	−5.8%
25% surface clamped	3.66	−11.4%

same boundary conditions would shift the frequency by approximately 33 %—but once the cavity is fluid-filled, the added mass appears to swamp much of the boundary stiffening effect. Clamping part of the shell forces the incompressible fluid to redistribute through a smaller aperture, increasing the effective modal mass. When this kinetic-energy increase exceeds the additional potential energy from the constraint, the net effect is a frequency decrease. The illustrative 6 % shift is well within the ±50–100% uncertainty band set by material property variability (§3), suggesting that the free-shell model is an adequate first approximation for the present purposes.

3.6. Oblate Spheroid Correction

The equivalent-sphere approximation used in Section 2 cannot capture the full geometric effect of an oblate shell. To gauge the size and sign of that effect, we also solved an exploratory Rayleigh–Ritz model using oblate spheroidal basis functions, with proper account of the spatially varying curvature and oblate fluid-mass geometry. This comparison must be interpreted cautiously: the same 2-DOF single-mode Ritz formulation lies about 9 % below the analytical sphere result even in the sphere limit ($c/a \to 1$ for the canonical $n = 2$ mode), owing to the restricted trial-function space. The 4–17% differences in Table 6 therefore should not be read as a precise oblate correction. Rather, they are exploratory and indicate the direction and approximate magnitude of geometric effects: for the canonical oblate shape, the more detailed model tends to reduce the flexural frequencies, with the reduction becoming more pronounced at higher eccentricity and higher E, where bending stiffness (which depends on local curvature) contributes a greater share of the restoring force.

Taken indicatively, the exploratory oblate Ritz model moves the canonical $n = 2$ frequency from 4.0 Hz to approximately 3.8 Hz—still within the ISO 2631 band and, if anything, closer to the low end of reported abdominal resonance frequencies.

Table 6: Illustrative equivalent-sphere vs. oblate spheroid Ritz frequencies for $c/a = 0.67$.

	$E = 0.1\,\mathrm{MPa}$		$E = 0.5\,\mathrm{MPa}$	
Mode	Sphere (Hz)	Ritz (Hz)	Sphere (Hz)	Ritz (Hz)
$n = 2$	4.0	3.8	7.1	6.2
$n = 3$	6.3	5.8	10.0	8.5
$n = 4$	8.9	8.1	12.9	11.3

3.7. Comparison with ISO 2631 Data

Table 7 compares model predictions against published abdominal resonance data. This comparison serves as an independent consistency check: the model, derived from first principles with no fitted parameters, predicts resonant frequencies consistent with empirical whole-body vibration data compiled over decades of occupational health research. Agreement with ISO 2631 therefore provides confidence that the shell model captures the essential physics of abdominal resonance.

Table 7: Model predictions vs. published abdominal resonance frequencies.

Body type	f_2 (Hz)	ISO 2631 range?
Lean ($E = 0.2$, $a = 15$ cm)	6.2	Yes
Average ($E = 0.1$, $a = 18$ cm)	4.0	Yes
Large ($E = 0.08$, $a = 20$ cm)	3.3	Marginal

4. Airborne vs. Mechanical Coupling

Having established that the abdomen possesses flexural resonances squarely in the infrasonic range, we now ask whether airborne sound can actually *excite* them. Under realistic airborne exposure conditions, the answer is effectively no.

4.1. Airborne Acoustic Pathway

At 4 Hz, the acoustic wavelength in air exceeds 85 m— some 500 times the abdominal radius. In this regime the abdomen is acoustically compact. Table 8 quantifies the resulting weak coupling. The energy-consistent displacement in column 3 is derived from the absorption cross-section (Eq. 29 below); the reciprocity argument is developed fully in §4.4. Modern analyses of scattering by pressurised elastic shells reach the same qualitative conclusion: long-wavelength radiation and absorption by low-order shell modes are strongly suppressed [32].

Table 8: Energy-consistent airborne displacement at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, canonical parameters). Displacements scale linearly with incident pressure; intermediate values may be interpolated. The pressure-based upper-bound estimate overestimates by a factor of ≈ 6.5 (ratio of ξ_{pressure} to ξ_{energy}).

SPL (dB)	p_{inc} (Pa)	ξ_{energy} (μm)	ξ_{pressure} upper bound (μm)	Energy-consistent $> 1\,\mu$m?
120	20.0	0.028	0.18	No
151	710	0.99	6.4	Nearly

At the $n = 2$ resonance, the energy-consistent airborne transfer function is

$$H_{\mathrm{air}}(\omega_2) = \frac{\xi}{p_{\mathrm{inc}}} \approx 1.4 \times 10^{-9}\,\mathrm{m/Pa}. \tag{24}$$

The displacements in Table 8 therefore scale linearly with incident pressure.

At 120 dB SPL—a level exceeding the conventional threshold of discomfort, and roughly the level at which conversation becomes impossible—the energy-consistent wall displacement is 0.028 µm. This falls two orders of magnitude below the 0.5–2.0 µm cellular activation range of mechanosensitive PIEZO1 and PIEZO2 channels [30], a benchmark derived from cell-culture experiments that provides a useful order-of-magnitude reference scale for biological mechanosensitivity. Reaching the 1 µm cellular benchmark on this energy-consistent basis demands approximately 151 dB (cf. 135 dB using the pressure-based upper-bound displacement, which overestimates by $\approx 6.5\times$), a level associated with severe acoustic hazards and available only from explosive blasts and rocket

launches. Modern mechanobiology strengthens rather than weakens that conclusion: PIEZO1 gating is tuned by membrane tension in the low-mN/m range, with half-maximal activation near 2 mN/m [33]. While the membrane-tension thresholds reported in cell-culture experiments cannot be directly translated to tissue-scale displacements, the order-of-magnitude comparison remains informative. Recent reviews summarise the broader physiological roles of PIEZO channels (following the 2021 Nobel Prize, and with structural detail from cryo-EM [34]) [35]. Of particular relevance here, PIEZO2-dependent gut mechanosensation is now known to regulate gastrointestinal transit [36]. The relevant question is therefore not whether the gut can sense deformation—it plainly can—but whether airborne infrasound can generate enough deformation. At such levels, auditory injury would be the more immediate clinical hazard than gastrointestinal effects.

4.2. Mechanical (Whole-Body Vibration) Pathway

Mechanical vibration, by contrast, bypasses the acoustic impedance barrier entirely. Vibration transmitted through the skeleton couples directly to the abdominal wall, bypassing the $(ka)^n$ penalty entirely. Because the bone–peritoneum impedance contrast is orders of magnitude smaller than the air–tissue mismatch, an upper-bound base-excitation model that assumes rigid skeletal transmission captures the essential physics [37, 38]. Table 9 shows substantially stronger coupling.

Table 9: Mechanical coupling at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, $\Gamma_2 = 1$). The EU Directive 2002/44/EC action value is $0.5\,\mathrm{m/s^2}$. Displacements scale linearly with base acceleration; intermediate values may be interpolated. The corrected values (§4.3) use $\Gamma_2 = 0.48$.

a_{rms} (m/s^2)	x_{base} (μm)	ξ_{rel} (μm)	ξ_{corr} (μm)†	Exceeds 1 µm?
0.1	229	917	440	Yes
0.5 (EU action)	1147	4586	2201	Yes

†Corrected for modal participation: $\xi_{\mathrm{corr}} = \Gamma_2\,\xi_{\mathrm{rel}}$ with $\Gamma_2 = 0.48$.

At the same resonance, the corresponding SDOF mechanical transfer function is

$$H_{\mathrm{mech}}(\omega_2) = \frac{\xi}{a_{\mathrm{rms}}} \approx 9.17 \times 10^{-3}\,\mathrm{m/(m\,s^{-2})}. \tag{25}$$

before the modal-participation correction discussed below.

At the EU action value of $0.5\,\mathrm{m/s^2}$, the predicted displacement exceeds the PIEZO cellular benchmark by more than three orders of magnitude. Unlike the airborne case, this is not a theoretical curiosity: it is consistent with occupational exposures among operators of heavy equipment, long-haul drivers, and helicopter pilots, in whom gastrointestinal complaints are well documented [2]. Recent seated-body biodynamic studies likewise show strong low-frequency nonlinearity, cross-axis coupling, and identifiable abdominally relevant modes in this same band [39, 40].

4.3. The Coupling Ratio

The system property of interest is the pair of transfer functions in Eqs. (24) and (25), not a single universal dimensionless number. For any chosen airborne pressure and mechanical acceleration, the corresponding displacement ratio is

$$\mathcal{R}(\omega) = \frac{H_{\mathrm{mech}}(\omega)\, a_{\mathrm{rms}}}{H_{\mathrm{air}}(\omega)\, p_{\mathrm{inc}}}. \tag{26}$$

Equation (26) is therefore pair-specific: the dimensionless ratio depends on the selected exposure levels. At the $n = 2$ resonance, the transfer-function ratio itself,

$$\frac{H_{\mathrm{mech}}}{H_{\mathrm{air}}} \approx 6.5 \times 10^{6}\,\mathrm{Pa\,s^2/m},$$

is a property of the shell–fluid system. Using $p_{\mathrm{inc}} = 20\,\mathrm{Pa}$ (120 dB SPL) and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ gives the headline SDOF upper bound

$$\mathcal{R}_{\mathrm{full}} \approx \frac{9.17 \times 10^{-3} \times 0.1}{1.4 \times 10^{-9} \times 20} \approx 3.3 \times 10^{4}.$$

Equivalently, to produce the same abdominal-wall displacement as 120 dB SPL airborne exposure, the required mechanical acceleration is only

$$a_{\mathrm{eq}} = \frac{H_{\mathrm{air}}\, p_{\mathrm{inc}}}{H_{\mathrm{mech}}} \approx 3.1 \times 10^{-6}\,\mathrm{m/s^2}, \tag{27}$$

which is 5.2 orders of magnitude below the EU occupational action value of $0.5\,\mathrm{m/s^2}$. The precise dimensionless ratio changes with the chosen reference pressure and acceleration, but the qualitative ordering—mechanical coupling far exceeding airborne coupling—is independent of that choice because it rests on the large transfer-function ratio $H_{\mathrm{mech}}/H_{\mathrm{air}}$. The ratio carries units of $\mathrm{Pa\,s^2/m}$; obtaining a dimensionless comparison at any frequency requires choosing a reference pressure and acceleration, but the enormous gap persists for all realistic exposure pairings.

Modal participation.. Vertical base excitation displaces the constrained portion rigidly, imposing a velocity boundary condition on the free shell at $\theta = \theta_c$. Figure 6 defines the corresponding geometry. The mechanical displacement in Eq. (26) assumes perfect coupling of vertical base excitation into the $n = 2$ mode (modal participation factor $\Gamma_2 = 1$). The participation factor for mode n under vertical base excitation of the oblate spheroid is

$$\Gamma_n = \frac{a \left| \int_{\cos\theta_c}^{1} \eta\, P_n(\eta)\, \mathrm{d}\eta \right|}{\int_{-1}^{1} P_n^2(\eta)\, \sigma(\eta)\, \mathrm{d}\eta}, \tag{28}$$

where $\eta = \cos\theta$, P_n is the Legendre polynomial of degree n, θ_c is the polar angle below which the shell is rigidly constrained by the pelvis and spine, and $\sigma(\eta) = \sqrt{c^2 + (a^2 - c^2)\,\eta^2}$ is the oblate-spheroid surface metric factor. The numerator integral quantifies the overlap between the vertical base-displacement pattern ($\propto \cos\theta = \eta$) and the mode shape $P_n(\eta)$ over the unconstrained cap; the denominator normalises that overlap by the corresponding modal mass on the full oblate surface. In this convention, Γ_n measures how efficiently rigid vertical support motion projects onto the nth shell mode.

For a free sphere ($\theta_c = \pi$), the numerator reduces to $\int_{-1}^{1} \eta\, P_n(\eta)\, \mathrm{d}\eta$, which vanishes for $n \neq 1$ by Legendre orthogonality; a free sphere therefore has $\Gamma_2 = 0$ exactly. At the opposite extreme, $\theta_c \to 0$ leaves no unconstrained cap, so the overlap integral also vanishes. Partial skeletal constraint breaks the free-sphere orthogonality and yields finite coupling into the $n = 2$ mode.

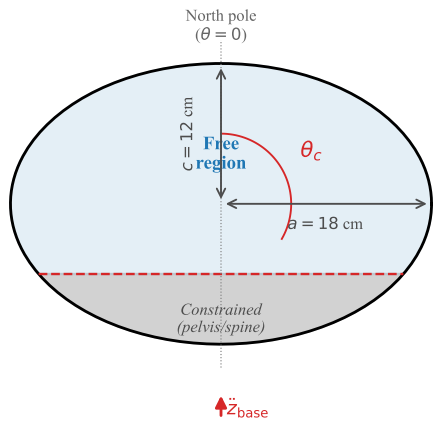

Figure 6: Constraint geometry for vertical base excitation. The inferior portion of the shell is rigidly supported below the polar angle θ_c, representing pelvic and spinal attachment, while the superior cap remains free to deform.

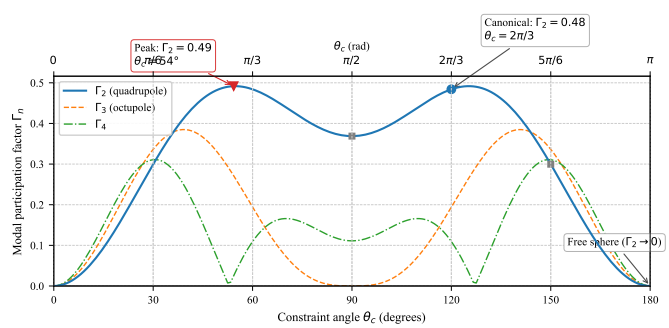

Figure 7: Modal participation factor Γ_2 as a function of constraint angle θ_c for the canonical oblate shell ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$). The factor vanishes at both extremes: fully clamped ($\theta_c = 0$) and free ($\theta_c = \pi$). The canonical pelvic constraint ($\theta_c = 2\pi/3$, dashed line) gives $\Gamma_2 \approx 0.48$.

The participation factor is non-monotonic in the constraint angle (Fig. 7): it vanishes at both extremes (fully clamped, $\theta_c \to 0$, and free, $\theta_c = \pi$) and reaches a maximum $\Gamma_2 \approx 0.49$ near $\theta_c \approx 125°$. At the canonical pelvic constraint ($\theta_c = 2\pi/3$), $\Gamma_2 \approx 0.48$, close to the peak; at $\theta_c = \pi/2$ (lower half constrained), $\Gamma_2 \approx 0.37$. Including this correction reduces the mechanical displacement to about 48% of the SDOF value and the coupling ratio to $\sim 1.6 \times 10^4$. The qualitative conclusion—a four-order-of-magnitude gap—is unaffected, but the uncorrected $\mathcal{R}_{\text{full}}$ reported above should be understood as an upper bound corresponding to the SDOF idealisation.

4.4. Energy Budget

We verify self-consistency through the absorption cross-section derived from reciprocity [17, 32]. The radiation efficiency of low-order shell modes is vanishingly small when $ka \ll 1$ [37]; formally:

$$\sigma_{\text{abs}} = \frac{(2n+1)\lambda^2}{4\pi} \frac{4\,\zeta_{\text{rad}}\,\zeta_{\text{struct}}}{(\zeta_{\text{rad}} + \zeta_{\text{struct}})^2}, \tag{29}$$

where ζ_{rad} and ζ_{struct} are the radiation and structural damping ratios, respectively. For the $n = 2$ mode in air, $\zeta_{\text{rad}} \approx 10^{-15}$ whereas $\zeta_{\text{struct}} = 0.125$, giving a Breit–Wigner absorption efficiency $4\zeta_{\text{rad}}\zeta_{\text{struct}}/\zeta_{\text{total}}^2 \sim 3 \times 10^{-14}$. The shell absorbs roughly three parts in a hundred trillion of the incident acoustic energy.

The energy-consistent displacement follows directly. The absorbed acoustic power is $\Pi_{\text{abs}} = \sigma_{\text{abs}}\, I$, where $I = p_{\text{inc}}^2/(2\rho_{\text{air}} c_{\text{air}})$ is the free-field intensity. At steady-state resonance, the time-averaged dissipated power is $\langle P_{\text{diss}} \rangle = c_{\text{eq}} \langle \dot{x}^2 \rangle$, where the equivalent viscous damping coefficient is $c_{\text{eq}} = 2\zeta\omega_n M_{\text{eff}}$ and the mean-square velocity of a harmonic oscillation with amplitude ξ is $\langle \dot{x}^2 \rangle = \omega_n^2 \xi^2/2$. Hence $\langle P_{\text{diss}} \rangle = \zeta\omega_n^3 M_{\text{eff}} \xi^2$. Equating absorbed and dissipated power gives

$$\xi_{\text{energy}} = \sqrt{\frac{\sigma_{\text{abs}}\, I}{\zeta_{\text{struct}}\, \omega_n^3\, M_{\text{eff}}}}, \tag{30}$$

where $M_{\text{eff}} = m_{\text{eff}}^{(n)} \times 4\pi R^2$ is the total modal mass. The $(2n+1)$ partial-wave degeneracy in $\sigma_{\max}$ and the corresponding $1/(2n+1)$ modal-mass normalisation cancel for the single azimuthal order excited by a plane wave. The factor of

$4\zeta_{\mathrm{rad}}\zeta_{\mathrm{struct}}/\zeta_{\mathrm{total}}^2 \sim 3 \times 10^{-14}$ inside σ_{abs} is the physical origin of the $\approx 6.5\times$ reduction from the pressure-based upper bound (equation 17): the pressure-based calculation implicitly assumes that all incident energy is available to drive the mode, whereas in fact only the Breit–Wigner fraction $4\zeta_{\mathrm{rad}}\zeta_{\mathrm{struct}}/\zeta_{\mathrm{total}}^2$ is absorbed. The pressure-based upper bound (with unity prefactor) exceeds the energy-consistent result by approximately $6.5\times$; with the exact multipole coefficient of 1/3 (§2.4), the pressure-based estimate reduces to approximately $0.06\,\mu$m, still a factor of ~ 2 above the energy-consistent value. This radiation inefficiency in the Rayleigh regime ($ka \ll 1$) is a well-established result in structural acoustics [17, 37].

4.5. Comparison with Empirical Evidence

The model predictions align with two otherwise contradictory bodies of evidence:

- **WBV studies**: ISO 2631 documents gastrointestinal symptoms at 4–8 Hz under mechanical vibration [1]. Our model explains *why* this frequency range is problematic.
- **Airborne infrasound studies**: No gastrointestinal effects have been observed at SPL ≤ 150 dB, although modern controlled exposure studies do report subtle neural effects without evidence of visceral resonance [8, 16]. Our model explains why gastrointestinal effects should not be expected.

The transfer-function disparity resolves this apparent contradiction (Figure 8): both literatures concern the same abdominal resonance, but through excitation pathways with markedly different coupling efficiencies.

5. Discussion

5.1. The Coupling Disparity

Figure 9 summarises this contrast schematically. The central result of this study is the separation between the airborne and mechanical transfer functions,

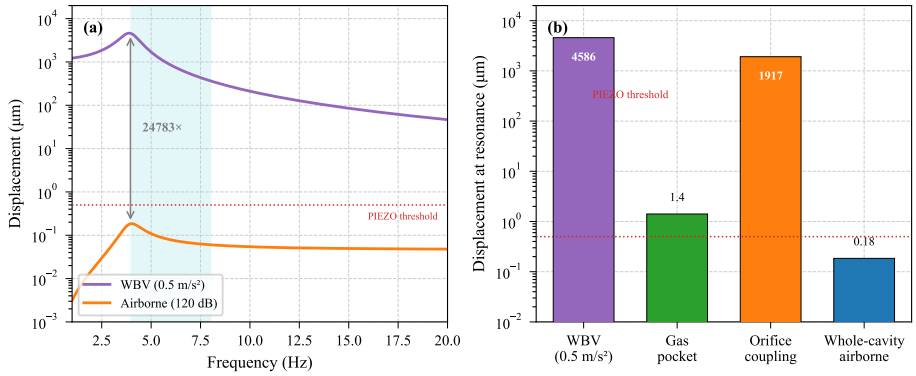

Figure 8: The central result. Left: wall displacement vs. frequency for airborne excitation at 120 dB SPL (lower curve) and mechanical whole-body vibration at $0.5\,\mathrm{m/s^2}$ (upper curve), showing a gap of roughly four orders of magnitude at resonance. Right: comparison of all four coupling pathways at the $n = 2$ resonance; the horizontal dashed line marks the PIEZO cellular mechanosensory reference scale.

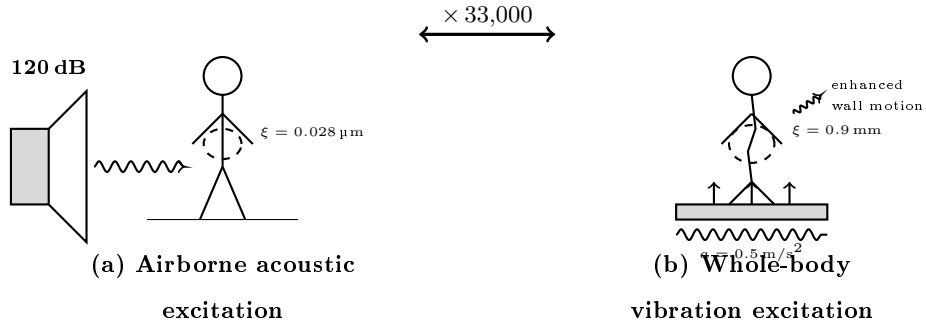

Figure 9: Schematic comparison of the two excitation pathways. (a) Airborne infrasound at 120 dB SPL produces wall displacements of 0.028 µm, well below cellular mechanotransduction benchmarks. (b) Whole-body vibration at occupational levels produces displacements roughly 3.3×10^4 times larger and readily exceeds those benchmarks. Schematic not to scale.

not any single exposure-pair ratio. Canonically, $H_{\mathrm{mech}}/H_{\mathrm{air}} \approx 6.5 \times 10^6\,\mathrm{Pa\,s^2/m}$ before the modal-participation correction, and Monte Carlo analysis shows that this transfer-function contrast remains decisive across physiologically plausible parameter ranges. For the commonly quoted reference pair (120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration), the same transfer-function separation implies the SDOF upper-bound displacement ratio $\mathcal{R}_{\mathrm{full}} \approx 3.3 \times 10^4$; including the canonical participation factor $\Gamma_2 \approx 0.48$ reduces this to $\sim 1.6 \times 10^4$ (Tables 8–9). The physical origin of this disparity lies in two independent mechanisms: a geometric filtering penalty that suppresses modal-pressure content, and an impedance mismatch that limits energy transfer.

First, the air–tissue impedance mismatch is severe. The specific acoustic impedance of air is $Z_{\mathrm{air}} = \rho_{\mathrm{air}} c_{\mathrm{air}} \approx 420\,\mathrm{Pa\,s/m}$, while that of soft tissue is $Z_{\mathrm{tissue}} \approx 1.6 \times 10^6\,\mathrm{Pa\,s/m}$, giving an intensity transmission coefficient $T_I = 4 Z_{\mathrm{air}} Z_{\mathrm{tissue}}/(Z_{\mathrm{air}} + Z_{\mathrm{tissue}})^2 \approx 1.1 \times 10^{-3}$. At the body surface, approximately 99.9 % of the incident acoustic power is reflected, before the still smaller mode-specific resonant absorption is accounted for. This general suppression of transmission across large biological impedance contrasts is consistent with modern frequency-dependent analyses of bone–soft tissue interfaces [38].

Second, even the transmitted pressure must couple to the spatial pattern of a flexural mode. In the long-wavelength (Rayleigh) limit, $ka \ll 1$, the n-th order component of the incident plane wave scales as $(ka)^n$ [17]. At the n=2 resonance ($f_2 \approx 4\,\mathrm{Hz}$) with $R_{\mathrm{eq}} = 0.157\,\mathrm{m}$, we obtain $ka = 2\pi f R/c_{\mathrm{air}} \approx 0.0114$ and therefore $(ka)^2 \approx 1.3 \times 10^{-4}$ for the $n = 2$ flexural mode (equation 16). Physically, the incident wavelength ($\lambda \approx 86\,\mathrm{m}$) is nearly 270 times the body diameter; the pressure field is almost perfectly uniform over the body surface, and the quadrupolar ($n = 2$) pressure gradient that could drive an oblate–prolate deformation is vanishingly small.

The energy budget analysis (§4) confirms this picture from an independent perspective. The radiation damping ratio in air is $\zeta_{\mathrm{rad}} \approx 10^{-15}$, whereas the structural damping ratio is $\zeta_{\mathrm{struct}} = 0.125$, so the absorption efficiency $\zeta_{\mathrm{rad}}/\zeta_{\mathrm{total}} \sim 10^{-14}$, giving a Breit–Wigner absorption efficiency $4\zeta_{\mathrm{rad}}\zeta_{\mathrm{struct}}/\zeta_{\mathrm{total}}^2 \sim$

3×10^{-14}. The shell is, for practical purposes, acoustically transparent at infrasonic frequencies. By contrast, the upper-bound mechanical model assumes rigid skeletal transmission: the bone–peritoneum impedance contrast is orders of magnitude smaller than the air–tissue mismatch, and the base-excitation drives the $n = 2$ mode via equation 19 with a modal participation factor $\Gamma_2 \approx 0.48$ (§4.3).

This disparity resolves an apparent paradox in the literature. Whole-body vibration at 4 Hz to 8 Hz is well documented to cause gastrointestinal distress—including nausea, abdominal pain, and in extreme cases, incontinence [2, 3, 41, 42, 43]—whereas airborne infrasound studies at levels up to 150 dB report no comparable effects [44, 8]. The present modal analysis shows that both observations are consistent with the *same* underlying resonance. The two excitation pathways differ not in the resonance they address, but in the efficiency with which they address it.

5.2. Implications for the Brown Note Hypothesis

The results permit a quantitative evaluation of the "brown note" hypothesis. The hypothesis contains one correct element: abdominal resonance in the 4 Hz to 10 Hz band is physically real, well-supported by the parametric study (§3), and consistent with ISO 2631 transmissibility data [1, 29]. Independent finite-element studies of the seated human body now recover dominant internal and abdominal vibration modes in the 4 Hz to 8 Hz range [45]. The flexural $n = 2$ mode at 4.0 Hz (for $E = 0.1$ MPa, relaxed musculature) falls squarely within the frequency band where occupational exposure studies report gastrointestinal symptoms.

Where the hypothesis fails is in the *excitation pathway.* At 120 dB SPL—a level exceeding the conventional threshold of discomfort and well above any environmental or entertainment exposure—the predicted airborne wall displacement is the energy-consistent value 0.028 µm (Table 8), roughly two orders of magnitude below the 0.5 µm to 2.0 µm cellular activation range of mechanosensitive PIEZO1 and PIEZO2 ion channels [30, 33, 35] (a benchmark derived

from cell-culture experiments; see Limitation 9). In the modern mechanotransduction literature, the more relevant gastrointestinal transducer is probably PIEZO2: somatosensory-neuron PIEZO2 signalling has now been shown to regulate gut transit directly [36]. Reaching the 1 µm cellular benchmark on the energy-consistent basis requires approximately 151 dB (Table 3 gives a median of 152 dB with a 90% credible interval of 145 dB to 156 dB), a level associated with severe acoustic hazards and encountered only in explosive blasts and rocket launches. Even at this extreme, the displacement remains three orders of magnitude below what routine whole-body vibration ($a_{\mathrm{rms}} = 0.5\,\mathrm{m/s^2}$) produces (Table 9).

The model also provides the first quantitative explanation for *why* whole-body vibration at 4 Hz to 8 Hz is so effective at producing gastrointestinal effects: it excites a flexural resonance of the fluid-filled abdominal cavity, producing millimetre-scale tissue displacements that far exceed cellular mechanotransduction benchmarks. The epidemiological pattern documented by ISO 2631—peak symptom incidence at 4 Hz to 8 Hz, with a strong dependence on acceleration amplitude—is precisely what a resonance-driven mechanism predicts.

5.3. Alternative Coupling Pathways

The whole-cavity resonance model of §2 treats the abdominal wall as a homogeneous fluid-filled shell. In reality, alternative local coupling mechanisms may also exist, notably compressible intestinal gas pockets and intermittent orifice coupling through the upper gastrointestinal tract. These mechanisms would operate through local compliance and transient boundary conditions rather than through the global $n = 2$ cavity resonance analysed here. We therefore treat them only as hypotheses for future study, not as co-equal conclusions of the present paper: the result established here is the severe weakness of airborne coupling to the global abdominal flexural mode relative to mechanical excitation. Assessing whether any local pathway can become biologically relevant under extreme conditions would require a dedicated model that resolves gas–tissue interactions, anatomical intermittency, and local mechanotransduction.

5.4. Clinical and Occupational Relevance

The transfer-function disparity has direct implications for occupational exposure standards. ISO 2631-1 specifies frequency-weighted acceleration limits for whole-body vibration, with the highest sensitivity weighting in the 4 Hz to 8 Hz band—precisely the range where our model predicts the $n = 2$ flexural resonance [1], and consistent with von Gierke and Brammer's compilation of organ resonance data [10]. More recent occupational reviews continue to link whole-body vibration exposure with adverse health outcomes across multiple organ systems [43]. The EU Physical Agents (Vibration) Directive 2002/44/EC codifies two thresholds: a daily exposure action value of $0.5\,\mathrm{m/s^2}$ $A(8)$ and an exposure limit value of $1.15\,\mathrm{m/s^2}$ $A(8)$. (The $A(8)$ action and limit values are frequency-weighted eight-hour equivalent accelerations; we use them here as illustrative equivalent harmonic amplitudes at the n=2 resonance frequency, not as direct predictions under the Directive's time-weighted assessment protocol.) At the action value, our model predicts a relative abdominal wall displacement of 4.6 mm at resonance (Table 9)—more than 4,000× the PIEZO cellular activation benchmark. The present analysis therefore provides a *mechanistic rationale* for why these particular frequency bands and amplitude thresholds were empirically selected: they correspond to conditions under which flexural resonance of the abdominal cavity produces tissue strains large enough to activate mechanotransduction pathways.

For airborne infrasound, ISO 7196 defines G-weighting but prescribes no dose–response limits. Our analysis supports this regulatory asymmetry: at environmental and occupational SPL ($\leq 120\,\mathrm{dB}$), airborne coupling to abdominal flexural modes is negligible, and exposure limits based on abdominal resonance effects are unnecessary at these levels.

A potentially valuable extension of this framework is toward personalised vibration exposure assessment. The parametric study (§3) shows that the $n = 2$ eigenfrequency depends sensitively on body habitus: larger abdominal radius (higher BMI) reduces the resonance frequency, while greater muscular tone (higher effective E) increases it. In principle, imaging-derived geometric and

viscoelastic parameters [26] combined with the analytical expressions of §2 could yield subject-specific resonance frequency estimates, enabling targeted vibration exposure limits for workers whose body habitus places their resonance frequency in a particularly hazardous band.

5.5. Broader Applications of the Analytical Framework

The brown note provided the motivation for this work, but the analytical framework—a fluid-filled viscoelastic shell, a modal decomposition into breathing and flexural families, and a systematic comparison of airborne versus mechanical coupling efficiency—is considerably more general. Any biological or engineered structure that can be idealised as a compliant enclosure surrounding a dense fluid falls within the scope of the model developed in §2. The two key dimensionless groups that govern the physics, the Helmholtz number ka and the impedance ratio $Z_{\text{wall}}/Z_{\text{fluid}}$, transfer directly; what changes between applications is the frequency range, the material properties, and the geometric aspect ratio of the shell. Potential applications include primary blast injury modelling—where ka approaches unity and the geometric filtering penalty largely disappears [46, 47]—swim-bladder target-strength prediction in fisheries acoustics [48, 49], and therapeutic ultrasound dosimetry where coupling must be maximised rather than shown to be negligible [50]. These extensions are beyond the scope of the present work but follow directly from the same modal decomposition and coupling-comparison methodology.

5.6. Nonlinear Amplitude Effects

The linear model of §2 is valid only for small displacements relative to the wall thickness. At the large amplitudes associated with whole-body vibration near the EU exposure limit, geometric nonlinearity from membrane stretching and fluid volume conservation may stiffen the response, further reducing actual displacement below the linear prediction. Such hardening behaviour is characteristic of thin-shell vibration at moderate amplitudes [22] and would shift the resonance peak to higher frequency while reducing its amplitude—effects

that are conservative for the brown note argument, since they widen the gap between prediction and any putative airborne effect. The airborne displacement ($\sim 0.028\,\mu\text{m}$) remains deep in the linear regime regardless. A detailed nonlinear analysis, including the amplitude-dependent frequency shift and hysteretic jump phenomena expected at occupational vibration levels, is deferred to future work.

5.7. Limitations

We identify the following limitations, grouped by their likely effect on the conclusions.

1. **Linear theory.** The Love–Kirchhoff shell formulation assumes small displacements. As discussed in §5.6, occupational vibration amplitudes likely exceed the linear regime, and the resulting hardening nonlinearity would reduce peak displacement below the linear prediction. Because the airborne displacement remains deep in the linear regime, nonlinearity *reduces* the coupling ratio somewhat but does not alter the qualitative conclusion.
2. **Homogeneous, isotropic wall.** The abdominal wall is a layered anisotropic composite comprising peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. The homogenisation error in effective bending stiffness is expected to be modest compared with inter-subject variability in wall thickness and elastic modulus. The isotropic assumption is a first-order approximation; anisotropy would split the mode degeneracies of the equivalent sphere but would not change the order of magnitude of the eigenfrequencies or the coupling ratio. Recent abdominal-wall mechanics reviews and elastography surveys place the canonical $E = 0.1\,\text{MPa}$ and $\eta = 0.25$ values as representative of contemporary measurement ranges, so the remaining uncertainty is quantitative rather than order-of-magnitude [23, 26].
3. **Free-shell boundary conditions.** The abdomen is partially constrained by the spine posteriorly, the pelvis inferiorly, and the rib cage superiorly.

The Rayleigh–Ritz study (§3.5) is best understood as an *exploratory sensitivity study*: it shows the range of possible frequencies under progressively constrained assumptions, from a free sphere ($f_2 = 4.13\,\mathrm{Hz}$) to 25% surface clamped ($f_2 = 3.66\,\mathrm{Hz}$, -11.4%). The basis-enrichment study in §3.5 shows monotone Ritz convergence from $3.7995\,\mathrm{Hz}$ to $3.6784\,\mathrm{Hz}$, so the reduced two-degree-of- freedom basis carries only a $3.3\,\%$ bias for the canonical $n = 2$ mode. These shifts therefore appear physically meaningful, while remaining illustrative because the constraint geometries are surrogate rather than anatomically exact. The entire bracketed range lies well within the material-property uncertainty band. Boundary conditions do, however, affect the mechanical pathway through the modal participation factor Γ_2 (§4.3), so the transfer-function separation remains large but is not literally unchanged.

4. **Thin-shell approximation.** The wall-thickness-to-radius ratio ($h/R \approx 0.06$) lies near the conventional upper bound of thin-shell theory ($h/R < 0.05$). First-order shear-deformation (Mindlin–Reissner) corrections would reduce the $n = 2$ frequency by approximately $4\,\%$ [51], a negligible effect compared with the material-property uncertainty. This limitation is unlikely to affect any conclusions.

5. **Equivalent-sphere geometry.** The oblate spheroidal abdomen is mapped to an equivalent sphere via $R_{\mathrm{eq}} = (a^2c)^{1/3}$. This preserves volume and leading-order curvature but suppresses the equatorial–polar mode splitting that occurs in an oblate spheroid [52]. For a typical aspect ratio $c/a = 0.67$, the splitting is estimated at $\sim$33 %, which could shift individual mode frequencies but would not change the coupling ratio or the qualitative separation between breathing and flexural mode families.

   More fundamentally, the human abdominal cavity is a *triaxial* ellipsoid (lateral $\sim$14 cm, craniocaudal $\sim$16 cm, anteroposterior $\sim$10 cm), not a spheroid of either kind. The oblate approximation captures the principal feature—anterior–posterior flattening—but any spheroid is a simplification. Crucially, however, the equivalent-sphere flexural frequencies depend on R_{eq}

alone: under the equivalent-sphere formula, at fixed R_{eq}, varying c/a from 0.3 (extremely oblate) to 3.0 (extremely prolate) changes f_2 by less than 1 %. The Ritz oblate-spheroid correction introduces additional aspect-ratio dependence (see Table 6), but the geometry enters the coupling ratio primarily through R_{eq}, so the qualitative results are insensitive to the choice of spheroidal idealisation.

6. **Inviscid fluid assumption.** The enclosed fluid is modelled as inviscid. A viscous correction using the Stokes boundary-layer thickness shows that even at the worst-case mucus viscosity ($\mu = 0.1\,\mathrm{Pa\,s}$), the quality factor decreases by only $\Delta Q = -1.85\%$. The crossover viscosity at which viscous damping would equal structural damping is $281\,\mathrm{Pa\,s}$, far above physiologically realistic abdominal fluid viscosities. The inviscid assumption is therefore well justified for any physiologically realistic abdominal fluid.
7. **No internal organ structure.** Visceral organs are homogenised into a single fluid. This estimate follows from a Hashin–Shtrikman upper-bound effective-medium calculation for a fluid matrix containing soft solid inclusions [53]: taking $K_{\mathrm{tissue}} \approx K_{\mathrm{water}}$, $\mu_{\mathrm{tissue}} \approx 1\,\mathrm{kPa}$, and a volume fraction $\phi = 0.25$ (representative of liver, spleen, and kidneys combined) predicts a frequency shift of only 0.4 % for the $n = 2$ mode. Because the organ tissue bulk modulus is close to that of water, the effective compressibility barely changes, and at sub-percolation volume fractions the isolated inclusions cannot transmit a system-spanning shear network. The homogeneous fluid model is therefore adequate for the present purposes.
8. **Speculative local-coupling hypotheses.** The brief discussion in §5.3 is intentionally schematic. Any realistic assessment of gas pockets or orifice coupling would require a coupled gas–tissue model with anatomical confinement, wall loading, and transient boundary conditions, all of which are omitted here.
9. **Mechanotransduction benchmarks.** The 0.5 µm to 2.0 µm PIEZO activation range [30, 33, 35] derives from cell-culture and membrane-tension experiments and may not directly translate to tissue-scale effects, where

strain concentrations around cells are modulated by the extracellular matrix. We therefore treat these values as cellular mechanosensory reference scales rather than experimentally confirmed tissue-scale thresholds. In gastrointestinal tissue the relevant transduction pathway is also likely to involve PIEZO2-expressing afferents rather than PIEZO1 alone [36]. This limitation introduces uncertainty in the absolute benchmark but does not affect the *relative* comparison between airborne and mechanical coupling.

In summary, the identified limitations affect the absolute values of predicted frequencies and displacements—in several cases substantially—but none alter the central finding that airborne acoustic coupling to abdominal flexural modes is 10^4–10^5 times weaker than mechanical coupling under occupationally relevant conditions.

5.8. Proposed Experimental Validation

A useful validation sequence begins with a tissue-mimicking phantom having the canonical spheroidal geometry and stiffness scale of §2. Modal testing of such a phantom would check the predicted flexural frequencies and mode shapes directly, while paired mechanical and airborne loading would test the central claim of this paper: that the coupling disparity between whole-body vibration and airborne infrasound is orders of magnitude rather than marginal. A natural in vivo extension would then combine subject-specific elastography with measured abdominal transmissibility, allowing the same framework to be assessed against human variability without changing the underlying model structure.

6. Conclusions

The central result of this study is the large separation between the airborne and mechanical transfer functions for the human abdominal cavity. At the canonical $n = 2$ resonance, the transfer-function ratio is $H_{\mathrm{mech}}/H_{\mathrm{air}} \approx 6.5 \times 10^6\,\mathrm{Pa\,s^2/m}$ under the SDOF idealisation before modal-participation correction. For the reference pair 120 dB SPL airborne loading and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$

RMS whole-body vibration, this system property implies a displacement ratio of approximately 3.3×10^4; including $\Gamma_2 \approx 0.48$ reduces this to $\sim 1.6 \times 10^4$. Derived from first-principles shell theory and energy-consistent reciprocity analysis, this resolves a long-standing empirical asymmetry: occupational whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects, while airborne infrasound at comparable frequencies and extreme intensities does not. More broadly, the same analysis provides a template for comparing excitation pathways in acoustically compact, fluid-loaded compliant structures, where geometry and impedance determine which forcing route can access a shared mode.

The disparity has a clean physical origin. The abdominal cavity possesses two distinct mode families: a breathing mode ($n = 0$) near 2500 Hz, governed by the fluid bulk modulus and therefore irrelevant at infrasonic frequencies; and flexural modes ($n \geq 2$) in the 4 Hz to 10 Hz range that match ISO 2631 abdominal transmissibility data. Airborne sound at these frequencies has a wavelength exceeding 85 m, placing the abdomen deep in the Rayleigh regime ($ka \approx 0.01$). The $(ka)^n$ geometric penalty suppresses modal-pressure content by four orders of magnitude; compounding this, the extremely low radiation efficiency at the air–tissue interface limits the energy available to drive the global resonance. An energy-consistent analysis confirms an absorption efficiency of $\sim 3 \times 10^{-14}$: the shell is, for practical purposes, acoustically transparent. Reaching the cellular mechanotransduction benchmark via airborne infrasound alone would require approximately 151 dB on the energy-consistent basis (135 dB using the less physical pressure-based upper-bound modal-displacement estimate)—a level at which structural damage to the middle ear and other severe acoustic hazards would present more immediate concerns than any gastrointestinal effects.

Mechanical vibration, transmitted through the skeleton and abdominal wall, couples directly and, for the reference comparison (120 dB SPL airborne versus $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS WBV with $\Gamma_2 = 1$), produces displacements approximately 3.3×10^4 times larger; including the participation correction leaves $\sim 1.6 \times 10^4$. This disparity is not a subtle biomechanical effect but a straightforward consequence of long-wavelength acoustics and impedance physics. Monte

Carlo uncertainty quantification on the same energy-consistent basis yields a reference-pair displacement ratio with median 1.3×10^4 and 90% credible interval $[2.4 \times 10^3,\ 4.6 \times 10^4]$; elastic modulus accounts for 86% of total variance.

The underlying resonance is genuine, but the airborne-coupling mechanism commonly invoked in brown-note claims is not supported. What popular culture attributes to airborne sound is more consistently explained by mechanical vibration—a distinction that matters considerably for occupational health standards. Airborne infrasound cannot excite the global cavity resonance to physiologically relevant amplitudes; whether localised compressible inclusions (intestinal gas pockets) provide a secondary transduction pathway at extreme intensities remains a hypothesis for future study, as such mechanisms would operate by local compliance rather than by the global cavity resonance established here.

Future work should verify the boundary condition assumptions through finite element analysis of anatomically realistic geometries, investigate the role of bowel gas pockets as localised acoustic transducers, and extend the model into the nonlinear regime relevant to high-amplitude mechanical excitation.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Formal Analysis, Software, Writing – Original Draft, Writing – Review & Editing. **Brian R. Mace**: Methodology, Validation, Writing – Review & Editing.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot (GitHub, Inc.; powered by large language models from OpenAI and Anthropic) in order to assist with: (1) implementing analytical expressions as Python code for numerical evaluation, (2) generating publication figures from computed results, (3) conducting structured literature searches, and (4) drafting and revising manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.